Question				Marking details	Marks Available					
					Total	AO1	AO2	AO3	Maths	Prac
12.	(a)			$\frac{1.44}{18.02}$ = 0.080 mol (1) e = $\frac{0.080}{0.020}$ = 4 (1)	2		1	1	1	
	(b)			$\frac{\frac{490}{1000}}{24.5}$ = 0.020 (1) c = $\frac{0.020}{0.020}$ = 1 (1)	2		1	1	1	
	(c)			white part of precipitate caused by $Mg(OH)_2$ produced from Mg^{2+} ions (1)	3	1				1
				grey-green precipitate so X is chromium/Cr (1)				1		1
				precipitate dissolves because Cr^{3+} is amphoteric (1)		1				1
	(d)			moles of HCl added = $\frac{25}{1000}$ × 1.00 = 0.025 mol (1)	4		1			1
				moles of NaOH required to neutralise remaining acid = $\frac{14}{1000}$ × 0.500 = 0.007 mol				1 1	1 1	1
				moles of HCl remaining = 0.007 mol (1)				1	1	
				moles of HCl reacting with OH^- and CO_3^{2-} = 0.025 – 0.007 = 0.018 mol (1)						
				Moles of HCl reacting with OH^- = 0.018–0.002 = 0.016 mol						
				d = $\frac{0.016}{0.001}$ = 16 (1)						
				ecf possible from part (b) for final mark						

Question				Marking details	Marks Available					
					Total	AO1	AO2	AO3	Maths	Prac
	(e)			award (3) for correct answer ⇨ $Mg_6Cr_2(CO_3)(OH)_{16}.4H_2O$ if correct answer not given award (1) for transferring identity of X and all numbers from parts (a)-(d) award (1) for balanced charges award (1) for 56 atoms present	3			1 1 1		
					14	**2**	**3**	**9**	**5**	**5**